probability is proportional to the number of atoms in your body. There's no reasonable justification for doing so, but who's to stop me? The only "rational" thing to do, according to this attitude, is to admit that there's no right way to assign credences, and therefore refuse to do so.

That is a position one is allowed to take, but I don't think it's the best one. If Many-Worlds is correct, we are going to find ourselves in situations of self-locating uncertainty whether we like it or not. And if our goal is to come up with the best scientific understanding of the world, that understanding will necessarily involve an assignment of credences in these situations. After all, part of science is predicting what will be observed, even if only probabilistically. If there were an arbitrary collection of ways to assign credences, and each of them seemed just as reasonable as the other, we would be stuck. But if the structure of the theory points unmistakably to one particular way to assign such credences, and that way is in agreement with our experimental data, we should adopt it, congratulate ourselves on a job well done, and move on to other problems.

Let's say we buy into the idea that there could be a clearly best way to assign credences when we don't know which branch of the wave function we're on. Before, we mentioned that, at heart, the Born rule is just Pythagoras's theorem in action. Now we can be a little more careful and explain why that's the rational way to think about credences in the presence of self-locating uncertainty.

This is an important question, because if we didn't already know about the Born rule, we might think that amplitudes are completely irrelevant to probabilities. When you go from one branch to two, for example, why not just assign equal probability to each, since they're two separate universes? It's easy to show that this idea, known as *branch counting*, can't possibly work. But there's a more restricted version, which says that we should assign equal probabilities to branches *when they have the same amplitude*. And that, wonderfully, turns out to be all we need to show that when branches have different amplitudes, we should use the Born rule.

Let's first dispatch the wrong idea of branch counting before turning to the strategy that actually works. Consider a single electron whose vertical